

Lab 3: Component Modelling & Architectural Pattern Selection

Problem Statement #32

Personal Subscription & Recurring Expense Auditor

NAME: SHREYA N HEGDE

SRN: PES1UG24AM401

Problem Overview:

A personal finance micro-app that aggregates recurring bank charges, alerts users before subscription renewal dates, calculates monthly burn rates, and provides one-click cancellation guides.

Actors:

- Individual User
- Finance Auditor

Architectural Style Analysis

Architecture	Advantages	Disadvantages	Suitability
Layered	Clear separation of concerns, easy maintenance	Some dependency between layers	High
Microservices	Independent scaling, fault isolation	More complexity and network overhead	Medium
Client-Server	Simple and centralized	Scalability and single-point-of-failure issues	Medium

Selected Architecture: Layered Architecture

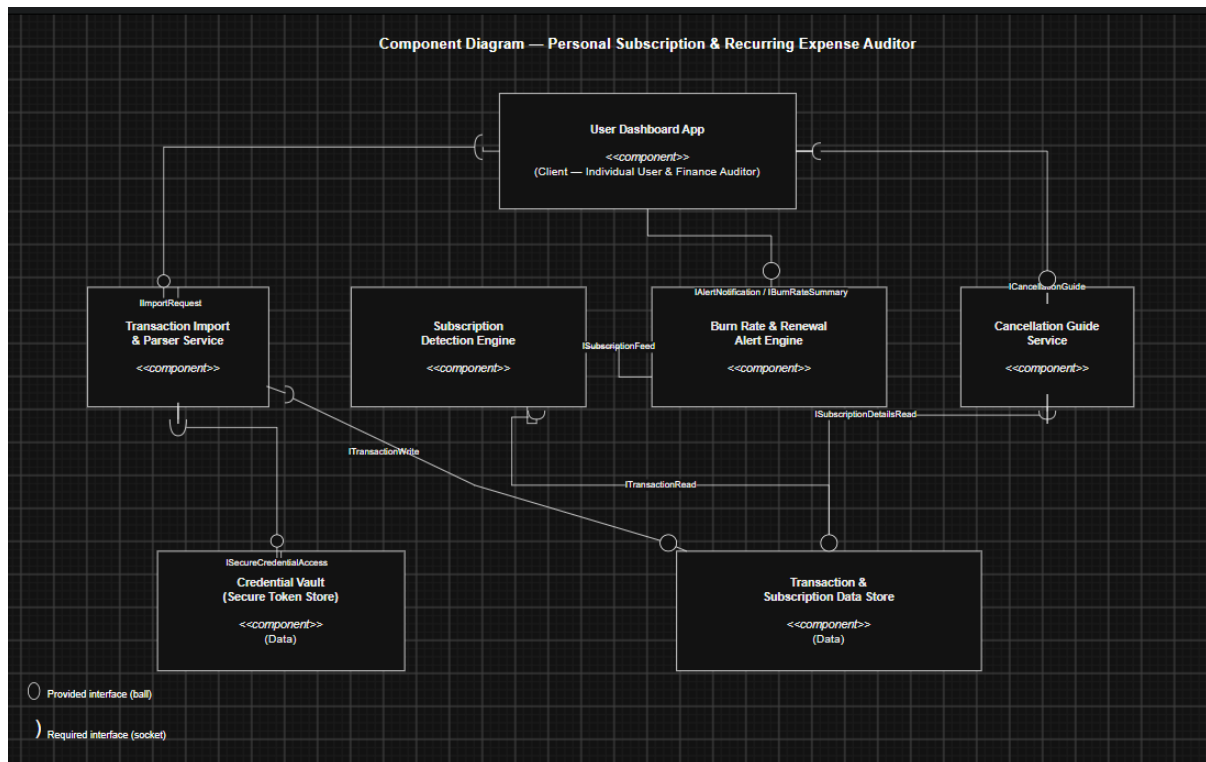
Reason: The application has clearly separated responsibilities such as user interaction, transaction analysis, subscription management, notifications, and data storage. Layered architecture provides a simple and maintainable structure for these functions.

Component Identification

System Components are:

1. User Interface
 - Displays subscriptions, burn rate and renewal alerts
 - Accepts user requests
2. Transaction Import & Analyzer
 - Imports transaction data
 - Identifies recurring billing patterns
3. Subscription Manager
 - Maintains subscription details
 - Calculates/maintains monthly subscription expenditure
 - Tracks renewal dates
4. Alert & Cancellation Manager
 - Generates renewal alerts
 - Provides cancellation guides
5. Finance Database
 - Stores transaction data
 - Stores subscription details
 - Stores renewal dates and expenditure information

MAIN COMPONENT DIAGRAM



Architectural Justification

Architecture Selection

We chose **Layered Architecture** for the Personal Subscription & Recurring Expense Auditor.

Reason 1 – Separation of Concerns

The architecture separates user interaction, transaction analysis, subscription management, notification handling, and data storage into distinct components. This makes the system easier to understand and maintain.

Reason 2 – Suitability

The application performs well-defined operations such as identifying recurring transactions, tracking renewal dates, calculating monthly expenditure, and providing cancellation guidance. Layered architecture provides a straightforward structure for these operations without the additional complexity of a distributed architecture.

Security Advantage

Financial transaction information can be isolated in the data layer and accessed through controlled application components rather than exposing the database directly to the user interface.

Performance Advantage

The business components can process subscription and expenditure information before presenting it to the user, reducing unnecessary processing at the interface level and improving response time.

These are exactly the categories the handout asks the justification to cover: architectural choice, two reasons, security advantage, and performance benefit.